Catching Bad Surrogate Models Without the Answer

The anchor-symmetry error certificate · 8 datasets · summary

Brandon Rodriguez · 16 October 2026

0	**8**	**3% → 100%**	**93%**
over-claims when the allowed range was right	datasets (lab, 4 wind, 2 solar, PDE)	wind beyond rated: range right, with published limit	error proven (K-S3, unseen 2021)

Every rule was fixed before its confirmation data were opened. Raw data from Zenodo, OEDI and OpenOA; code at github.com/rdgbrandon/ad_model_prediction.

The certificate: three terms

Full certificate

$$\| f_\theta(x) - f(x) \|_2 \geq S(x) = \max_{\lambda \in \Lambda_{\text{anc}}(x)} \frac{\left[\sum_{j=1}^{3} r_j(x, \lambda) \right]_+}{\|A_\lambda\|_2}$$

$r_1 = D_\theta(x, \lambda)$	$r_2 = -b_\lambda$	$r_3 = -\beta_\lambda(x)$	$[u]_+ = \max(u, 0)$	$\Lambda_{\text{anc}}(x)$
Defect	Minus anchor allowance	Minus law allowance	Clip at zero	Admitted anchors (one here)

The guaranteed minimum error: the defect minus both allowances, if both allowances hold.

Law allowance

$$\beta_\lambda(x) = \kappa' \min\left(\left[\ln \frac{x}{x_{\text{law}}^{\max}} \right]_+, \ \delta^* \right) + r_\lambda$$

$$\kappa = \max_h \frac{\|\bar{y}(n) - \hat{c}_h(n)\|_2}{d_h}, \qquad \delta^* = \max_j \frac{\ln y_{\text{cap},j} - \bar{y}_j(a^*)}{q_j}, \qquad r_\lambda = \|f(T_\lambda x) - c(T_\lambda x)\|_2$$

$\kappa' = 2\kappa$	$[\cdot]_+$	δ^*	r_λ	$\beta_\lambda(x)$
Declared growth rate	Clip negatives to zero	Distance where law hits cap	Law's measured miss at anchor	Law departure bound

Grows with the log-distance past the calibration ceiling, and stops growing at the published limit.

$S(x) = [D - b - \beta]_+$. If the allowances hold, the model's true error is at least S.

The datasets

Dataset	Machines	Units tested	Predicts	Published limit
Capillary waves (lab)	1 apparatus	20 trials	12-band spectrum	none
Kelmarsh wind	6 turbines	72 turbine-months / yr	power, rotor speed	2,050 kW · 15 rpm
Penmanshiel wind	5 turbines	60	power, rotor speed	2,050 kW · 17.1 rpm
La Haute Borne wind	4 turbines	93	power	2,050 kW
Hill of Towie wind	21 turbines	252	power	2,300 kW
Solar PVDAQ 9068	24 combiners	384	DC current	on sunlight only
Solar PVDAQ 2107	24 inverters	1,671	AC power	27.6 kW
Neural PDE surrogates	36 networks	1,200 test points	PDE solution	exact symmetry

High range hidden in every dataset: models trained low, certificate computed without the hidden data, hidden data used only to check.

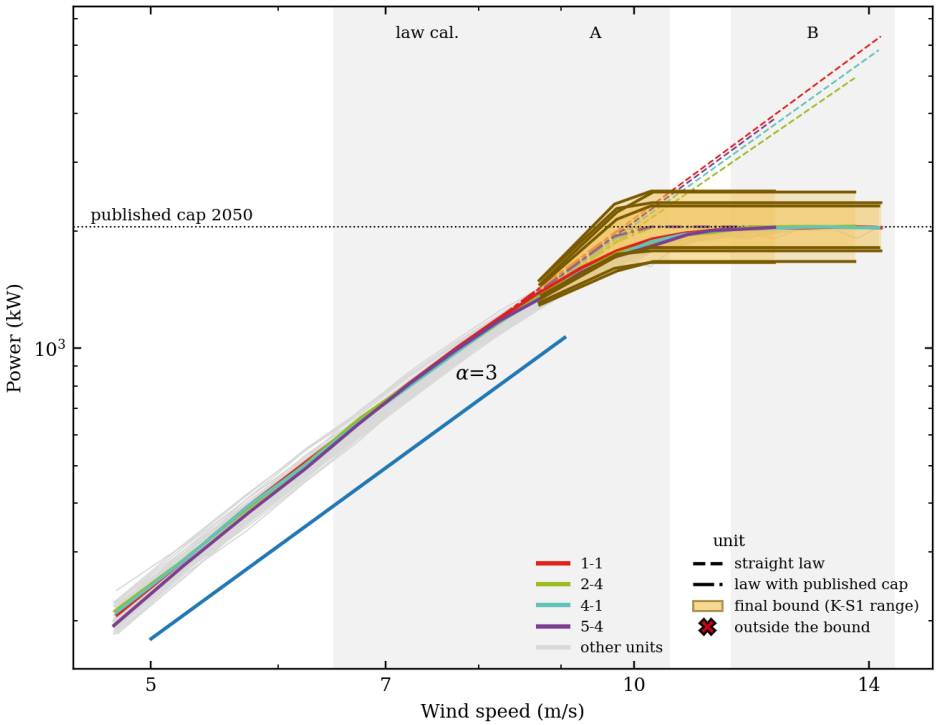
Results at a glance (beyond the published limit)

Dataset	Straight law: range right	With limit: range right	Bound valid	Error proven
Kelmarsh wind	3%	100%	100%	0.80
Penmanshiel wind	82%	100%	100%	0.62
Hill of Towie wind	65%	100%	100%	0.72
La Haute Borne wind	100%	100%	100%	0.28
Solar 2107	95%	98%	100%	0.53
Solar 9068	99%	99%	100%	0.02
Capillary waves	100%	100%	100%	0.52

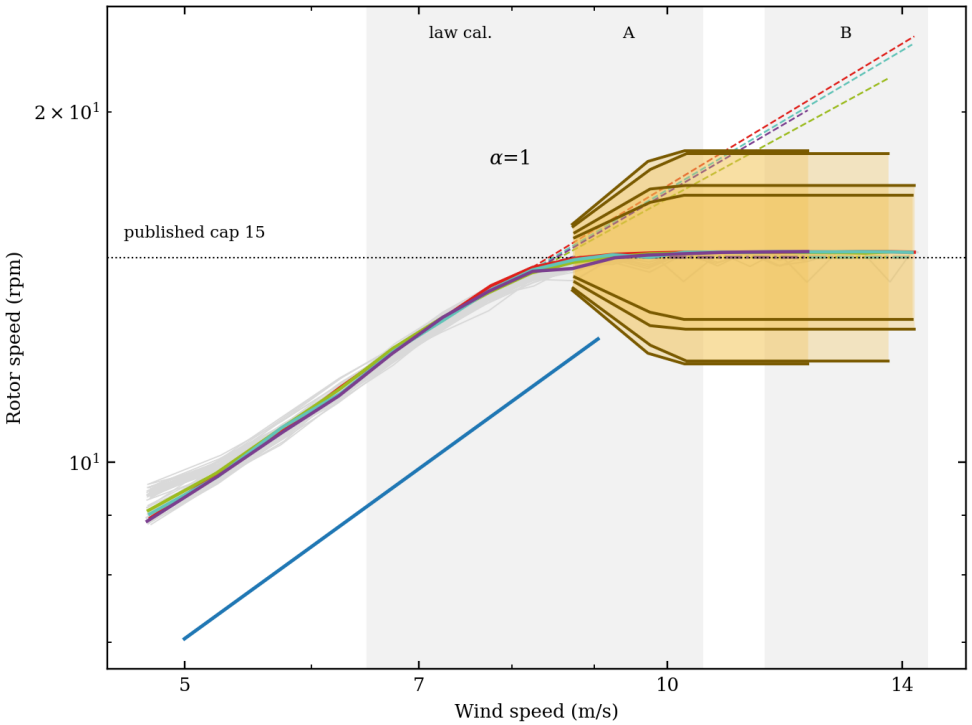
Range right = the law's true error was inside the allowance. Error proven = median share of the true error the certificate guarantees.

What the allowed range looks like (Kelmarsh wind)

(a) Kelmarsh wind 2020: power

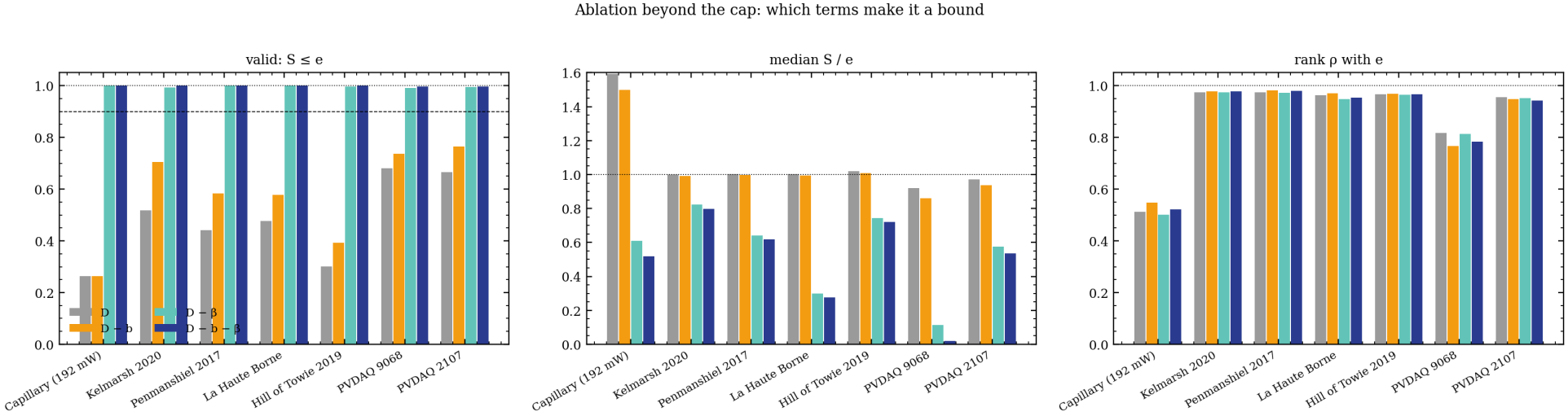


(b) Kelmarsh wind 2020: rotor speed



Gold band = allowed range from physics + the published limit. Red × = measured point outside it. One slide per dataset in the full deck.

Only the full three-term sum is a bound

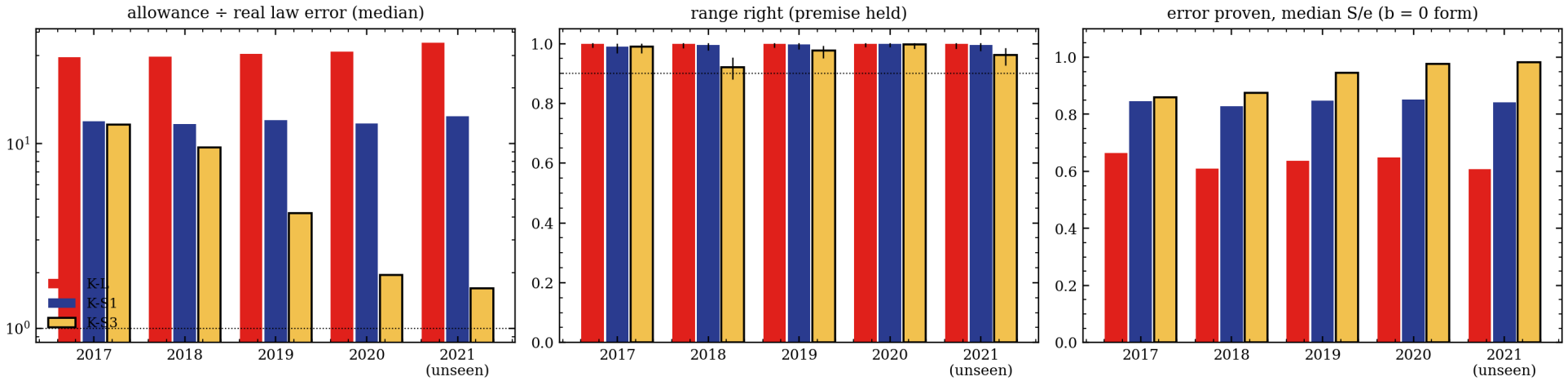


D alone is valid on 26–68% of rows; D − β on 94–100%; the full certificate D − b − β on 98–100%.

Tighter past the limit: K-S3 (Kelmarsh 2021, unseen)

14.1× → 1.6×	96%	99%	0.77 → 0.93
allowance ÷ real law error	range right	bound valid	error proven

Kelmarsh, beyond rated power: shrinking the allowance after the cap (K-S3)



Rule fixed on 2017–2020; 2021 downloaded afterwards. Gold = K-S3.

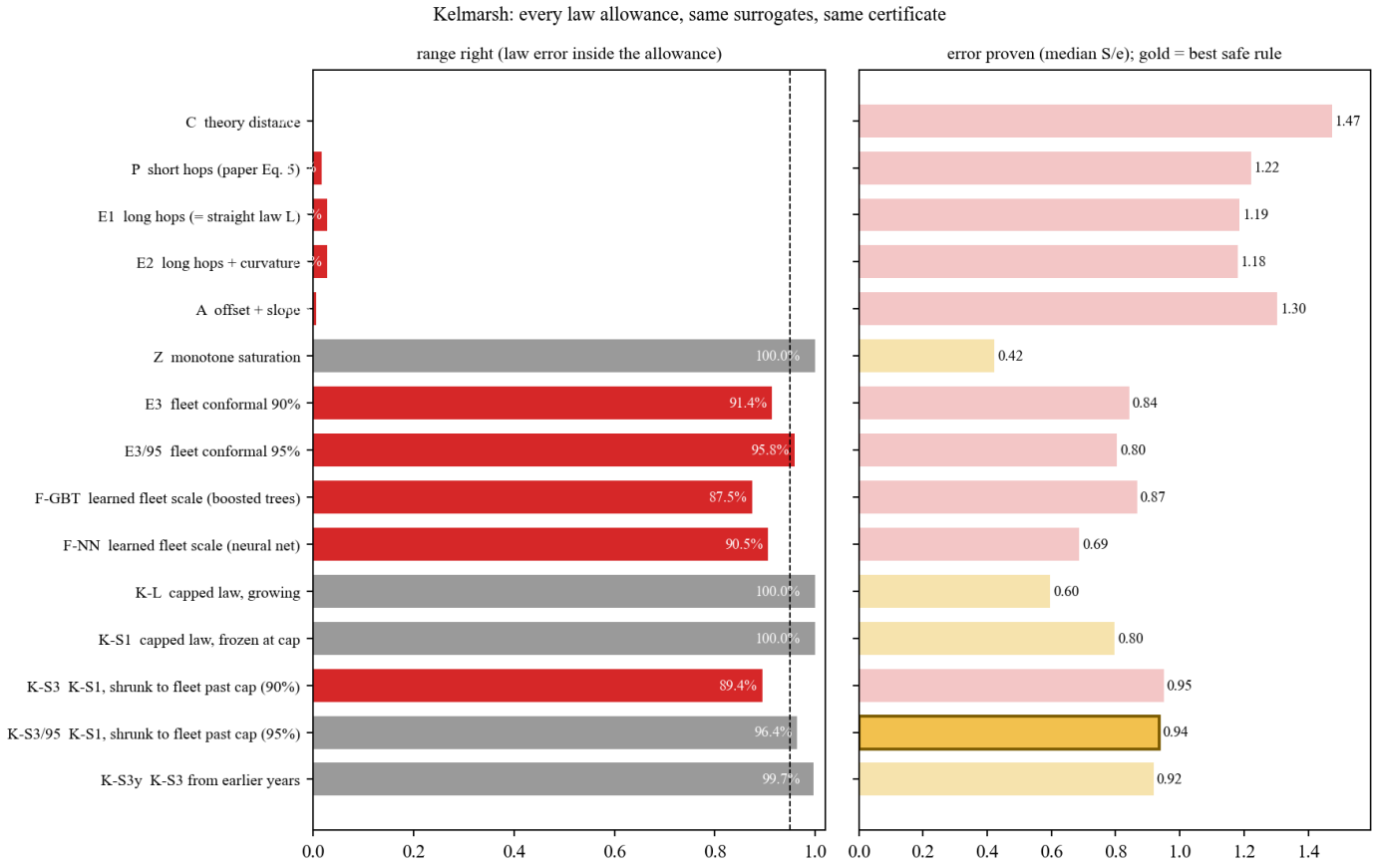
Every law allowance on every dataset



Same units, 9 surrogates, same three-term certificate; only β changes. Fleet rules use other machines in other months. Boxed = best rule with range right ≥ 95%.

All methods: the takeaway

0	13	K-S3	K-S1 / Z
bound failures where the range was right	law allowances × 7 datasets	best when fleet history exists (0.94, 0.77)	best guaranteed rules otherwise



Single-unit rules (C, P, E1, A) are unsafe beyond the limit. Fleet rules prove the most but sit at 87–97% range right.

Winner per dataset

Dataset	Best safe rule	Range right	Bound valid	Error proven	K-S1 (before)
Kelmarsh	**K-S3/95**	96%	99%	**0.94**	0.80
Penmanshiel	**K-S1**	100%	100%	**0.62**	0.62
Hill of Towie	**K-S1**	100%	100%	**0.72**	0.72
La Haute Borne	**Z**	100%	100%	**0.29**	0.28
Solar 2107	**K-S3/95**	97%	99%	**0.77**	0.53
Solar 9068	**Z**	96%	100%	**0.23**	0.02
Capillary	**C**	100%	100%	**0.73**	0.52

Safe = range right ≥ 95% and bound valid ≥ 99%. Picked after seeing results (exploratory). Capillary has no published limit: last column is the paper rule P.

Real neural surrogates: PDE solvers

Exact symmetry law (PDE surrogates)

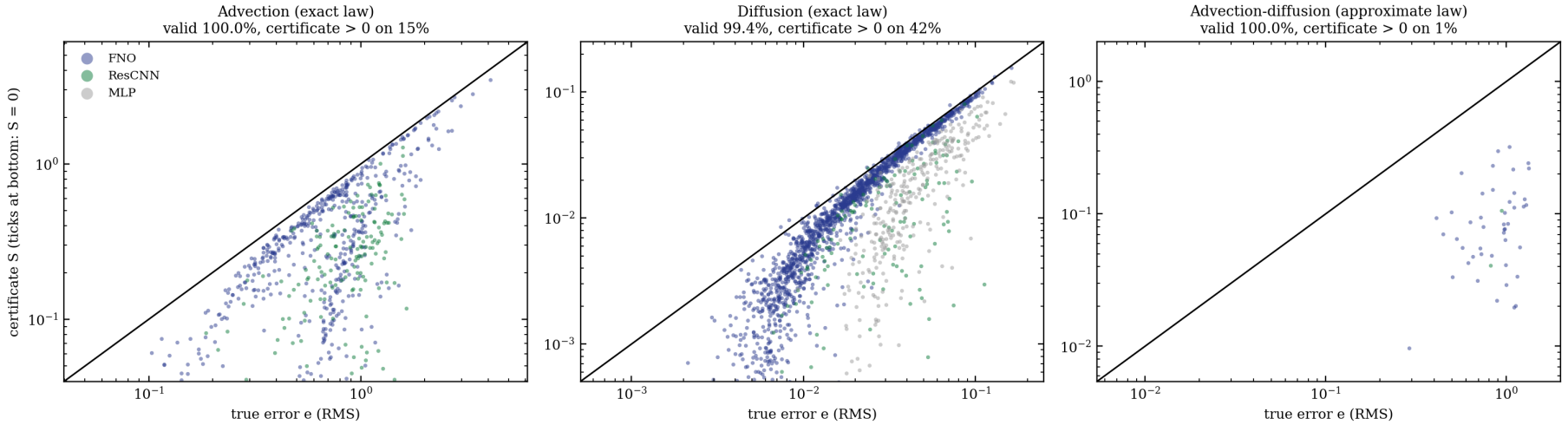
$$u(x,\,t;\,p) = u(x,\,\lambda t;\,p/\lambda) \quad \Rightarrow \quad T_\lambda(u_0, p, t) = (u_0,\; p/\lambda,\; \lambda t),\;\; A_\lambda = I,\;\; c_\lambda = 0$$

p	t	λ	u_0	$p\,t$
speed b or diffusivity D	time	transform parameter	initial condition	invariant (unchanged by T)

Advection and diffusion depend on p and t only through p·t, so an out-of-range p maps to an in-range anchor.

99–100%	**63%**	**0%**	**36%–63%**
bound valid (all systems)	error proven when it speaks (diffusion)	Lipschitz bound says something	conformal covers out of range

Neural PDE surrogates beyond their training range: every point below the diagonal is a valid bound



36 networks (FNO, CNN, MLP) trained in range, tested beyond it; fresh test set. Below the diagonal = valid.

Which law allowance to use

Your system	Use	Result
Published limit + machine history	**K-S3**	96–99.7% right, proves 0.92–0.94 (Kelmarsh)
Published limit + a fleet, no history	**K-S3 at 95%**	93–97% right, proves 0.77–0.94
Published output limit only	**K-S1**	99.7–100% right, proves 0.28–0.80
Output can only flatten, no limit	**Z**	96–100% right, proves 0.23–0.68
Exact symmetry (PDE)	**$\beta = 0$**	99–100% valid
Unknown bend, one system	**none**	add theory, simulation or data

Unsafe beyond the limit: theory distance (C), short hops (P), long hops (E1), offset + slope (A). Learned fleet scales add nothing over E3/K-S3.

Is it publishable?

Venue	Ready?	Needs
Workshop (ML for physical sciences, UQ)	**Yes, now**	—
Applied journal (data-centric engineering, wind energy)	**Close**	one realistic wind ML model; per-site results
Top ML conference	**Not yet**	fix silent PDE cases; numeric comparison to set-membership bounds

Strong	**Modest**	**Practical**
rigour: rules fixed first, unseen confirmation	novelty: distance-to-set bound, applied	win: published limits make it reliable

Main references: Milanese & Novara 2004 (set membership); Calliess 2016 (Lipschitz); Vovk et al. 2005 (conformal); Li et al. 2021 (FNO); Takamoto et al. 2022 (PDEBench).